

Lec 12

Recap

- ① CDF properties $\rightarrow F_X(x) = P(X \leq x)$
- ② Expectation $\rightarrow E[X] = \sum_{x \in \mathcal{X}_0} P(x) x$
- ③ Variance

$$\rightarrow \text{Var}(X) = E[(X - E[X])^2]$$

Scaling & Shifting

$$E[aX + b] = aE[X] + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Non-negative X , $E[X] = 0 \Rightarrow X$ is 0 w.p. 1

$$\text{Var}(X) = 0 \Rightarrow X = E[X] \text{ w.p. 1}$$

- ① Monotonic
- ② Right Continuous
- ③ $\lim_{x \rightarrow -\infty} F_X(x) = 0, \lim_{x \rightarrow \infty} F_X(x) = 1$

Today

- ① Binomial, Bernoulli, Geometric & Poisson R.V.

- ② Joint PMF of multiple R.Vs.

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = E[(X - E[X])^2]$$

$$= \sum_{x \in \mathcal{X}_0} (x - E[X])^2 P_X(x)$$

$$= \sum_{x \in \mathcal{X}_0} (x^2 - 2xE[X] + (E[X])^2) P_X(x)$$

$$= \sum_{x \in \mathcal{X}_0} x^2 P_X(x) - 2 \left(\sum_{x \in \mathcal{X}_0} x P_X(x) \right) E[X] + (E[X])^2 \sum_{x \in \mathcal{X}_0} P_X(x)$$

$$= E[X^2] - 2E[X]E[X] + (E[X])^2$$


$$\underline{\text{Standard deviation}} = \sqrt{\text{Var}(X)}$$

Examples :

(a) $X \sim \text{Bernoulli}(p)$.

$$P_X(x) = \begin{cases} p & x=1 \\ 1-p & x=0 \end{cases}$$

$$E[X] = \sum_{x \in \{0,1\}} x P_X(x) = 0 \times (1-p) + 1 \times p = p$$

$$E[X^2] = \sum_{x \in \{0,1\}} x^2 P_X(x) = 0^2(1-p) + 1^2 p = p$$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= p - p^2 = p(1-p) \end{aligned}$$

(b) $X \sim \text{Binomial}(n, p)$ # of heads seen in
n-coin tosses

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad k \in \{0, 1, \dots, n\}$$

$$\begin{aligned} E[X] &= \sum_{k=0}^n k P_X(k) \\ &= n \sum_{k=1}^n \frac{(n-1)!}{(n-k)! (k-1)!} p^k (1-p)^{n-k} \end{aligned}$$

$$\begin{aligned}
 E[X] &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1-p)^{n-1-(k-1)} \\
 &= np \sum_{k'=0}^{n-1} \binom{n-1}{k'} p^{k'} (1-p)^{n-1-k'} \quad (k'=k-1) \\
 &= np (p + (1-p))^{n-1} = np.
 \end{aligned}$$

$$\begin{aligned}
 E[X^2] &= \sum_{k=1}^n \frac{(k^2-k) n!}{(n-k)! k!} p^k (1-p)^{n-k} \\
 &\quad + \sum_{k=1}^n k P_X(k) \\
 &= \sum_{k=2}^n \frac{p^2 n(n-1)(n-2)!}{(n-k)! (k-2)!} p^{k-2} (1-p)^{(n-2)-(k-2)} + E[X] \\
 &= n(n-1)p^2 [p + (1-p)]^{n-2} + E[X] \\
 E[X^2] &= n(n-1)p^2 + np
 \end{aligned}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\begin{aligned} \text{Var}(X) &= n(n-1)p^2 + np - \cancel{n^2 p^2} \\ &= np(1-p) \end{aligned}$$

$$\hat{X} \sim \text{Bernoulli}(p)$$

$$E[\hat{X}] = p$$

$$\text{Var}[\hat{X}] = p(1-p)$$

$X_1, X_2, \dots, X_n$ are Bernoulli(p).

$$X \sim \text{Binomial}(n, p)$$

$$X = X_1 + X_2 + \dots + X_n$$

Linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i]$$

Geometric(p)

$$\Omega = \{H, TH, TTH, \dots\}$$

$$P(\{\underbrace{TT}_{k-1} - H\}) = P(A_i) + P(A_i^c) = P(A_i)$$

Prob of seeing head at i th toss is equal to p .

Event A_i : seeing head at i th toss.

$$P(A_i) = p.$$

$$X(\omega) = \text{length of } \omega \quad \omega \in \Omega$$

$$H \rightarrow 1$$

$$TH \rightarrow 2$$

$$TTH \rightarrow 3$$

$\vdots$

$$P_X(k) = P(A_1^c \cap A_2^c \cap \dots \cap A_{k-1}^c \cap A_k) \\ = (1-p)^{k-1} p$$

$$\begin{aligned} \sum_{k=1}^{\infty} P_X(k) &= \sum_{k=1}^{\infty} (1-p)^{k-1} p \\ &= p(1 + (1-p) + (1-p)^2 + \dots) \\ &= p \left[\frac{1}{1-(1-p)} \right] = 1 \end{aligned}$$

$$1 + a + a^2 + \dots = \frac{1}{1-a}$$

Χωρίς προσήκη (p)

$$\begin{aligned} E[X] &= \sum_{k=1}^{\infty} k P_X(k) \\ &= \sum_{k=1}^{\infty} k (1-p)^{k-1} p \rightarrow \textcircled{2} \end{aligned}$$

$$S(x) = \sum_{k=1}^{\infty} (1-x)^k = \frac{1-x}{x} = \frac{1}{x} - 1$$

$$\begin{aligned} \frac{\partial S(x)}{\partial x} &= \sum_{k=1}^{\infty} -k (1-x)^{k-1} \\ &= -\frac{1}{x^2} \end{aligned}$$

$$\Rightarrow \sum_{k=1}^{\infty} k (1-x)^{k-1} = \frac{1}{x^2} \rightarrow \textcircled{1}$$

Substitute ① in ②

$$E[X] = p \times \frac{1}{p^2} = \frac{1}{p}$$

$$E[X^2] = \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1}p + E[X] \rightarrow (4)$$

$$\begin{aligned} \frac{\partial S^2(x)}{\partial x^2} &= \frac{2}{x^3} \\ &= \sum_{k=1}^{\infty} k(k-1)(1-x)^{k-2} \end{aligned}$$

$$\Rightarrow \sum_{k=1}^{\infty} k(k-1)(1-x)^{k-1} = \frac{2}{x^3}(1-x) \rightarrow (5)$$

Substitute (5) in (4)

$$\begin{aligned} E[X^2] &= p \times \frac{2}{p^3}(1-p) + \frac{1}{p} \\ &= \frac{2}{p^2}(1-p) + \frac{1}{p} = \frac{2}{p^2} - \frac{1}{p} \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - (E(X))^2 \\
 &= \sum \frac{x^2}{p^x} - \frac{1}{p} - \frac{1}{p^2} \\
 &= \frac{1}{p^2} - \frac{1}{p} \\
 &= \frac{(1-p)}{p^2}
 \end{aligned}$$

Poisson random Variable

$X \sim \text{Poisson}(\lambda)$

Used to model the following:

① $X \sim \text{Poisson}(\lambda)$ to model # packets incoming in time t

② # of successful trials in n trials when n is large

$X \sim \text{Binomial}(n, p)$ such that $np = \lambda$

As $n \rightarrow \infty$ i.e., $p \rightarrow 0$, $P_X(k) \rightarrow \frac{e^{-\lambda} \lambda^k}{k!}$

$$\begin{aligned} P_X(k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \frac{n!}{(n-k)! k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \end{aligned}$$

$$\lim_{n \rightarrow \infty} P_X(k) = \frac{n(n-1) \cdots (n-k+1)}{k!} \frac{\lambda^k}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} P_X(k) &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \frac{n(n-1) \cdots (n-k+1)}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
 &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} 1 \cdot \left(1 - \frac{\lambda}{n}\right) \left(1 - \frac{\lambda}{n}\right) \cdots \left(1 - \frac{\lambda}{n}\right) \lim_{n \rightarrow \infty} \frac{\left(1 - \frac{\lambda}{n}\right)^n}{\left(1 - \frac{\lambda}{n}\right)^k} \\
 &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \frac{\left(1 - \frac{\lambda}{n}\right)^n}{\left(1 - \frac{\lambda}{n}\right)^k} \\
 &= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \\
 &= \frac{\lambda^k}{k!} e^{-\lambda}
 \end{aligned}$$

$$\begin{aligned}
 a_n &\rightarrow a \\
 b_n &\rightarrow b
 \end{aligned}$$

$$\lim_{n \rightarrow \infty} a_n b_n \rightarrow ab$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$$

$$\lim_{n \rightarrow \infty} P_X(x) = \frac{\lambda^k}{k!} e^{-\lambda}$$

of Successful trials

= # of people who get matching hats

N people with N hats

Each person picks a hat without replacement

E_i = person i gets their own hat

$$P_N = P(E_1^c \cap E_2^c \dots E_N^c)$$

$$= 1 - P(E_1 \cup E_2 \dots E_N)$$

$$P(E_1 \cup E_2 \dots E_N) = \sum P(E_i) - \sum_{1 \leq i < j \leq N} P(E_i \cap E_j) + \dots$$

$$e^{-1} = \sum_{k=0}^{\infty} \frac{1^k}{k!}$$

$$P(E_i) = \frac{(N-1)!}{N!}$$

$$P(E_1 \cap \dots \cap E_k) = \frac{(N-k)!}{N!}$$



$$P(E_1 \cup \dots \cup E_N) = N \times \frac{1}{N} - \frac{N!}{(N-2)! \times 2!} \frac{(N-2)!}{N!} + \dots$$

$$= 1 - \frac{1}{2!} + \frac{1}{3!} - \dots + (-1)^{N-1} \frac{1}{N!}$$

$$P(E_1 \cup \dots \cup E_N) = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{1}{N!}$$

$$\rightarrow e^{-1}$$

$X_i = \# \text{ of Successful trials}$
 $= \# \text{ perfect hat-person matching}$

$$P_X(k) \rightarrow \frac{e^{-1}}{k!} \approx \text{PMF of Poisson}(1) \text{ R.V.}$$

$$P(X=0) = 1 - 1 + \frac{1}{2!} - \dots + \frac{1}{N!} \rightarrow e^{-1}$$

$$P(X=k) = \binom{N}{k} \frac{(N-k)!}{N!} P_{N-k} = \frac{P_{N-k}}{k!} \xrightarrow{N \rightarrow \infty} \frac{e^{-1}}{k!}$$

$$P(E_1, \neg E_2, \dots, \neg E_k, E_{k+1}^c, \neg E_N^c)$$

$$= P(E_1, \neg E_k) P(E_{k+1}^c, \neg E_N^c | \text{---})$$

$$= \frac{(N-k)!}{N!} P_{N-k}$$